

Auteur: Zaky Souai
 Studierichting: Informatica
 Oefeningenreeks 4A nr. 42.17

$$\int \frac{6x^2 + 7x + 5}{3x^2 + 7x + 2} dx$$

$$3x^2 + 7x + 2 = (3x + 1)(x + 2)$$

$$\frac{6x^2 + 7x + 5}{3x^2 + 7x + 2} = 2 + \frac{-7x + 1}{3x^2 + 7x + 2}$$

$$\frac{-7x + 1}{3x^2 + 7x + 2} = \frac{-7x + 1}{(3x + 1)(x + 2)} = \frac{A}{3x + 1} + \frac{B}{x + 2}$$

$$A = \frac{-7x + 1}{x + 2} \bigg|_{x = -\frac{1}{3}} = \frac{\frac{10}{3}}{\frac{3}{3}} = 2$$

$$B = \frac{-7x + 1}{3x + 1} \bigg|_{x = -2} = \frac{15}{-5} = -3$$

$$\frac{6x^2 + 7x + 5}{3x^2 + 7x + 2} = 2 + \frac{2}{3x + 1} - \frac{3}{x + 2}$$

$$\int \frac{6x^2 + 7x + 5}{3x^2 + 7x + 2} dx = \int 2dx + 2 \int \frac{dx}{3x + 1} - 3 \int \frac{dx}{x + 2}$$

$$= 2x + \frac{2}{3} \ln |3x + 1| - 3 \ln |x + 2| + C$$

References